

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

EC-302 (GS)
B.E. III Semester Examination, June 2020
Grading System (GS)
Computer System Organization
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Represent the following conditional control statement by two register transfer statements with control functions. If ($P = 1$) then ($R_1 \leftarrow R_2$) else if ($Q = 1$) then ($R_1 \leftarrow R_2$).
2. List and explain most common addressing techniques with advantages and disadvantages.
3. What are the steps in the execution of C program stored in memory? What are the possible ways to increase the program execution speed?
4. What are the different registers of a basic computer? Draw the block diagram of the hardware that implements the following register transfer statement.
 $yT_2 : R_2 \leftarrow R_1, R_1 \leftarrow R_2$
5. Write short notes on any two :
 - i) CPU
 - ii) RAM and ROM
 - iii) Dynamic and static RAM.
6. What is the difference between a direct and an indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into a processor register?

OR

- a) What is Cache Memory? Why is it used? Define cache hit and cache miss.
- b) Name three techniques of mapping. Explain any one of them.
7. How many 128×8 RAM chips are needed to provide a memory capacity of 2048 bytes? How many lines of the address bus must be used to access 2048 bytes of memory?

OR

What is the purpose of the system bus controller? Explain how the system can be designed to distinguish between references to local memory and references to common shared memory.

8. Answer any four of the following:
 - a) Explain Handshaking.
 - b) What is the principle of micro programming?
 - c) Distinguish Synchronous bus and Asynchronous bus.
 - d) Explain virtual memory.
 - e) What are the major characteristics of a pipeline?
 - f) Define microprocessors.